UC SAN DIEGO / ECE 251C

10
Orthogonal filter banks

Orthonormal coordinates, energy preservation, and conjugate mirror filters

When is a filter bank an energy-preserving change of coordinates?

10.1 Analysis filtering computes inner products

A perfect-reconstruction filter bank retains enough information to recover its input. An *orthogonal* filter bank does more: its outputs are coordinates in an orthonormal basis, so it also preserves energy. We now identify the basis vectors hidden inside the filtering and downsampling operations.

For $x \in \ell^2(\mathbb{Z})$, let $g_0, g_1 \in \ell^2(\mathbb{Z})$ be synthesis filters and set

$$h_\ell[n] = g_\ell^*[-n], \qquad \ell \in \{0, 1\}. \tag{10.1}$$

The asterisk denotes complex conjugation. We allow complex-valued filters and signals; the conjugation disappears when a filter is real. This is the zero-delay convention used in Lecture 8.

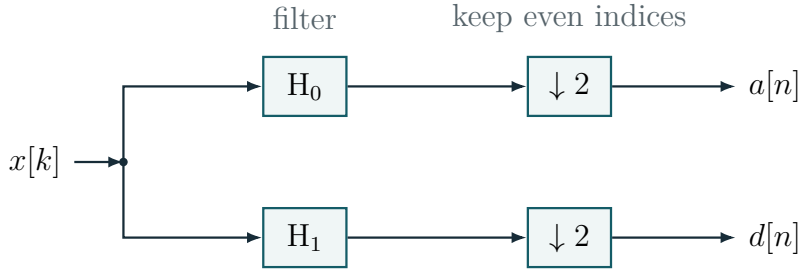


Figure 10.1: Filtering combines input samples. Downsampling evaluates the filtered sequence at $2n$; it does not itself perform averaging.

Using $(h * x)[r] = \sum_k x[k]h[r-k]$, the two outputs are

$$\begin{aligned} a[n] &= (h_0 * x)[2n] = \sum_k x[k] g_0^*[k-2n], \\ d[n] &= (h_1 * x)[2n] = \sum_k x[k] g_1^*[k-2n]. \end{aligned} \tag{10.2}$$

Indeed, $h_\ell[2n-k] = g_\ell^*[-(2n-k)] = g_\ell^*[k-2n]$. This change in the order of subtraction is what turns convolution into an inner product.

10.2 The basis vectors are even translates of two filters

Recall the inner product and its induced norm:

$$\langle x, y\rangle_{\ell^2} = \sum_{k\in\mathbb{Z}} x[k]y^*[k], \qquad \|x\|_{\ell^2}^2 = \langle x, x\rangle_{\ell^2}.$$

The *second* sequence is conjugated. Define

$$v_{\ell,n}[k] \coloneqq g_\ell[k - 2n]. \tag{10.3}$$

Then (10.2) says exactly that

$$a[n] = \langle x, v_{0,n}\rangle_{\ell^2}, \qquad d[n] = \langle x, v_{1,n}\rangle_{\ell^2}.$$

The index ℓ chooses a channel; n chooses a translate. The shift is $2n$, not n, because the filtered outputs are retained only at even indices.

Definition 10.1 (Orthogonal filter bank)**.** The bank with analysis filters (10.1) is *orthogonal* if

$$\mathcal{B} = \{v_{\ell,n} : \ell \in \{0, 1\},\ n \in \mathbb{Z}\}$$

is an orthonormal basis of $\ell^2(\mathbb{Z})$. This requires both pairwise orthonormality and completeness: finite linear combinations of these vectors must approximate every finite-energy sequence in the ℓ^2 norm.

Example: Haar. Let $g_0 = (1, 1)/\sqrt{2}$ and $g_1 = (1, -1)/\sqrt{2}$, supported at $0, 1$. At each n, their translates occupy the pair $\{2n, 2n + 1\}$. On that pair they are perpendicular unit vectors; distinct pairs do not overlap.

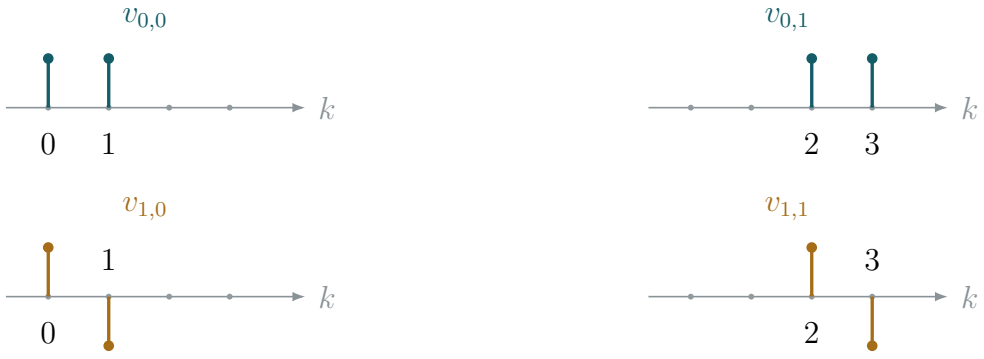


Figure 10.2: Two neighboring Haar pairs. Every nonzero tap has magnitude $1/\sqrt{2}$. Each pair contains an average direction and a difference direction.

The word *orthogonal* is conventional for the bank. Over complex sequences, the resulting change of coordinates is unitary.

10.3 Synthesis is the basis expansion

Suppose first that $\mathcal{B}$ is an orthonormal basis. The same coordinate rule used for Fourier series now gives

$$\begin{aligned} x[k] &= \sum_{\ell=0}^{1} \sum_{n\in\mathbb{Z}} \langle x, v_{\ell,n}\rangle_{\ell^2} v_{\ell,n}[k] \\ &= \sum_n a[n] g_0[k-2n] + \sum_n d[n] g_1[k-2n]. \end{aligned} \tag{10.4}$$

The series converges in ℓ^2. The summation index n selects a coefficient; k is the location at which we evaluate the reconstructed sequence.

Why upsampling comes before synthesis filtering. Define the zero-inserted sequence by

$$(\uparrow 2\, a)[r] = \begin{cases} a[r/2], & r \text{ even}, \\ 0, & r \text{ odd}. \end{cases}$$

Only $r = 2n$ contributes to its convolution with g_0, so

$$\big(g_0 * (\uparrow 2\, a)\big)[k] = \sum_r (\uparrow 2\, a)[r] g_0[k-r] = \sum_n a[n] g_0[k-2n].$$

The second term in (10.4) is obtained in the same way from d.

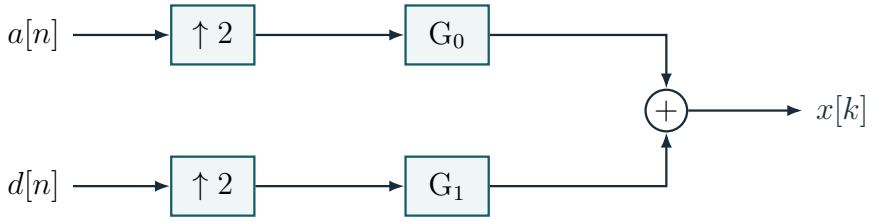


Figure 10.3: The synthesis bank is the basis expansion: place each coefficient on the even grid, filter with its basis generator, and add the two channels.

Energy is preserved. For any finite set F of basis indices, set $p_F = \sum_{(\ell,n)\in F} \langle x, v_{\ell,n}\rangle v_{\ell,n}$. Orthogonality makes the cross terms vanish, and gives

$$\|x - p_F\|_{\ell^2}^2 = \|x\|_{\ell^2}^2 - \sum_{(\ell,n)\in F} |\langle x, v_{\ell,n}\rangle_{\ell^2}|^2.$$

The vector p_F is the orthogonal projection onto the included basis vectors, so it is the best approximation from their span. Completeness supplies finite approximations with arbitrarily small error. Once F contains the vectors in such an approximation, $\|x - p_F\|_{\ell^2}$ is at least as small. Thus, as F exhausts the indices,

$$\boxed{\|x\|_{\ell^2}^2 = \|a\|_{\ell^2}^2 + \|d\|_{\ell^2}^2.} \tag{10.5}$$

10.4 A criterion using double-shift correlations

The definition involves infinitely many basis vectors. Translation lets us test them using only the two original filters. Define their *double-shift correlations* by

$$r_{\ell s}[k] \coloneqq \sum_{j\in\mathbb{Z}} g_\ell[j]g_s^*[j-2k]. \tag{10.6}$$

The factor two again comes from downsampling. These are correlations, not convolutions: the shifted sequence in the second factor is conjugated. For finite-energy filters, Cauchy–Schwarz ensures absolute convergence:

$$\sum_j |g_\ell[j]g_s^*[j-2k]| \le \|g_\ell\|_{\ell^2}\|g_s\|_{\ell^2} < \infty.$$

A shift does not change a sequence's ℓ^2 norm.

Theorem 10.2 (Double-shift orthogonality)**.** Let $g_0, g_1 \in \ell^2(\mathbb{Z})$ and $v_{\ell,n}[j] = g_\ell[j-2n]$. Then $\{v_{\ell,n}\}_{\ell\in\{0,1\},n\in\mathbb{Z}}$ is an orthonormal basis of $\ell^2(\mathbb{Z})$ if and only if

$$\boxed{r_{\ell s}[k] = \delta_{\ell s}\delta[k] \quad (\ell, s \in \{0,1\},\ k \in \mathbb{Z}).} \tag{10.7}$$

Here $\delta_{\ell s} = 1$ when the channel indices agree and zero otherwise. Explicitly,

$$r_{00}[k] = r_{11}[k] = \delta[k], \qquad r_{01}[k] = r_{10}[k] = 0.$$

Proof. **First establish orthonormality.** For any two channel/shift pairs,

$$\begin{aligned}\langle v_{\ell,n}, v_{s,m}\rangle_{\ell^2} &= \sum_j g_\ell[j-2n]g_s^*[j-2m] \\ &= \sum_u g_\ell[u]g_s^*[u-2(m-n)] \qquad\qquad (u = j-2n) \\ &= r_{\ell s}[m-n].\end{aligned}$$

Thus (10.7) is exactly pairwise orthonormality. It is necessary for an orthonormal basis, and it proves that our family is orthonormal when the conditions hold. We still have to prove *completeness*.

The cross conditions are not independent. Indeed, substituting $u = j + 2k$ gives

$$r_{01}^*[-k] = \sum_j g_0^*[j]g_1[j+2k] = \sum_u g_1[u]g_0^*[u-2k] = r_{10}[k].$$

It therefore suffices to check the two autocorrelations and one cross-correlation.

How much of a unit impulse is captured? Write $\delta_j[k] = \delta[k-j]$. Its energy is one, and

$$\langle \delta_j, v_{\ell,n} \rangle_{\ell^2} = \sum_k \delta[k-j] g_\ell^*[k-2n] = g_\ell^*[j-2n].$$

For $N \geq 0$, form the *finite* sum

$$p_{j,N} = \sum_{\ell=0}^{1} \sum_{n=-N}^{N} \langle \delta_j, v_{\ell,n} \rangle_{\ell^2} v_{\ell,n}.$$

We do not yet assume that $p_{j,N}$ converges to δ_j. That is the step to prove.

Put $c_{\ell,n} = \langle \delta_j, v_{\ell,n} \rangle_{\ell^2}$ for this calculation. Orthonormality gives

$$\begin{aligned} \|p_{j,N}\|_{\ell^2}^2 &= \sum_{\ell,s=0}^{1} \sum_{n,m=-N}^{N} c_{\ell,n} c_{s,m}^* \langle v_{\ell,n}, v_{s,m} \rangle_{\ell^2} \\ &= \sum_{\ell=0}^{1} \sum_{n=-N}^{N} |c_{\ell,n}|^2. \end{aligned}$$

Conjugating the coefficient in the second argument also gives

$$\langle \delta_j, p_{j,N} \rangle_{\ell^2} = \sum_{\ell=0}^{1} \sum_{n=-N}^{N} c_{\ell,n}^* \langle \delta_j, v_{\ell,n} \rangle_{\ell^2} = \sum_{\ell=0}^{1} \sum_{n=-N}^{N} |c_{\ell,n}|^2.$$

Expand the squared error:

$$\begin{aligned} \|\delta_j - p_{j,N}\|_{\ell^2}^2 &= \|\delta_j\|_{\ell^2}^2 - 2\,\mathrm{Re}\langle \delta_j, p_{j,N} \rangle_{\ell^2} + \|p_{j,N}\|_{\ell^2}^2 \\ &= 1 - \sum_{\ell=0}^{1} \sum_{n=-N}^{N} |g_\ell[j-2n]|^2. \end{aligned}$$

Since a squared norm is nonnegative, the finite sum cannot exceed one. Increasing N adds nonnegative terms, so

$$S_j := \sum_{\ell=0}^{1} \sum_{n \in \mathbb{Z}} |g_\ell[j-2n]|^2 \leq 1. \tag{10.8}$$

This is Bessel's inequality for the unit impulse, obtained directly. The number S_j is the energy captured from that impulse by all the filter translates. Completeness will follow if every S_j equals one.

The even and odd impulses leave no missing energy. For $j = 0$, the indices $-2n$ cover all even integers. For $j = 1$, the indices $1 - 2n$ cover all odd integers. Hence

$$S_0 = \sum_{\ell=0}^{1} \sum_{k \text{ even}} |g_\ell[k]|^2, \qquad S_1 = \sum_{\ell=0}^{1} \sum_{k \text{ odd}} |g_\ell[k]|^2.$$

Together they account for every tap of both filters exactly once:

$$S_0 + S_1 = \sum_{\ell=0}^{1} \sum_{k} |g_\ell[k]|^2 = \|g_0\|_{\ell^2}^2 + \|g_1\|_{\ell^2}^2 = 2.$$

The last equality is (10.7) at zero shift in each channel. But $S_0 \le 1$ and $S_1 \le 1$, so necessarily $S_0 = S_1 = 1$.

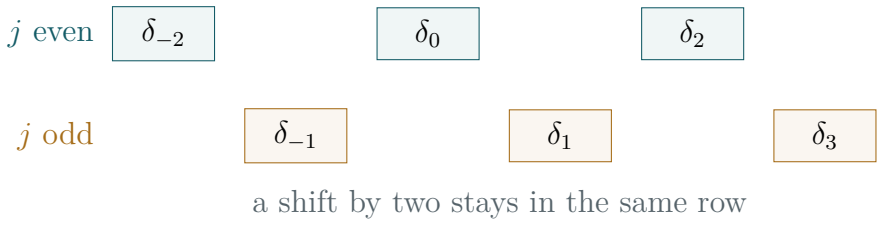


Figure 10.4: There are only two translation classes of unit impulses. The two unit-norm generators force the captured energy in each class to be exactly one.

For any $j = 2q + r$, with $r \in \{0, 1\}$, substitute $u = n - q$:

$$S_j = \sum_{\ell=0}^{1} \sum_{n} |g_\ell[2q + r - 2n]|^2 = \sum_{\ell=0}^{1} \sum_{u} |g_\ell[r - 2u]|^2 = S_r = 1.$$

The error formula on the preceding page now gives $\|\delta_j - p_{j,N}\|_{\ell^2} \to 0$. Every unit impulse belongs to the closed span of our family.

Finally, any $x \in \ell^2(\mathbb{Z})$ is approximated by its finite truncations:

$$x_M = \sum_{|j| \le M} x[j]\delta_j, \qquad \|x - x_M\|_{\ell^2}^2 = \sum_{|j| > M} |x[j]|^2 \longrightarrow 0.$$

Each x_M lies in that closed span, and therefore so does x. This proves completeness and finishes the theorem. □

The two-channel structure is essential to this argument. An arbitrary orthonormal family need not be complete. Here the even/odd counting proves that no sequence direction is missing.

10.5 Autocorrelation describes the power response

We next translate a condition on the filter taps into a condition on the frequency response. For the design calculations from here onward, the filters are FIR, so all sums involving their taps are finite.

To distinguish the full autocorrelation from its even samples, write

$$r[k] \coloneqq \sum_n g_0[n] g_0^*[n-k], \qquad r_{00}[m] = r[2m]. \tag{10.9}$$

The low-pass part of (10.7) therefore requires $r[2m] = \delta[m]$. It does *not* require $r[k] = 0$ at odd shifts.

Expand the squared magnitude. With the Fourier notation from Lecture 3,

$$\hat{g}_0(\omega) = G_0(\mathrm{e}^{\mathrm{j}\omega}) = \sum_n g_0[n] \mathrm{e}^{-\mathrm{j}\omega n}.$$

Complex conjugation changes both the coefficient and the sign of the exponential. Consequently,

$$\begin{aligned} |\hat{g}_0(\omega)|^2 &= \hat{g}_0(\omega)\hat{g}_0(\omega)^* \\ &= \left(\sum_n g_0[n]\mathrm{e}^{-\mathrm{j}\omega n}\right)\left(\sum_m g_0^*[m]\mathrm{e}^{\mathrm{j}\omega m}\right) \\ &= \sum_{n,m} g_0[n] g_0^*[m] \mathrm{e}^{-\mathrm{j}\omega(n-m)} \\ &= \sum_k \left(\sum_n g_0[n] g_0^*[n-k]\right) \mathrm{e}^{-\mathrm{j}\omega k} \qquad (k = n-m) \\ &= \sum_k r[k] \mathrm{e}^{-\mathrm{j}\omega k}. \end{aligned}$$

Thus the *power response* $|\hat{g}_0|^2$ is the DTFT of the full autocorrelation. The inverse DTFT recovers $r[k]$ with the opposite exponential sign:

$$r[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\hat{g}_0(\omega)|^2 \mathrm{e}^{\mathrm{j}k\omega} \, \mathrm{d}\omega.$$

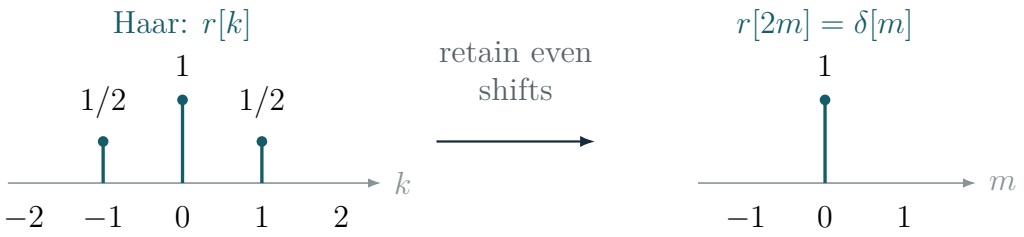


Figure 10.5: The Haar low-pass filter overlaps its one-sample translate, giving $r[\pm 1] = 1/2$. Only the even shifts are required to be orthogonal.

A half-period shift selects the even correlations. Since $\mathrm{e}^{-\mathrm{j}\pi k} = (-1)^k$,

$$\begin{aligned} |\widehat{g}_0(\omega)|^2 + |\widehat{g}_0(\omega+\pi)|^2 &= \sum_k r[k]\big(1+(-1)^k\big)\mathrm{e}^{-\mathrm{j}\omega k} \\ &= 2\sum_m r[2m]\mathrm{e}^{-\mathrm{j}2m\omega}. \end{aligned}$$

The factor in parentheses is two for even k and zero for odd k. If $r[2m] = \delta[m]$, the last sum is the constant two.

Conversely, suppose the sum of the two power responses is two for all ω. Divide the preceding identity by two, multiply by $\mathrm{e}^{\mathrm{j}2q\omega}$, and integrate:

$$\begin{aligned} \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{e}^{\mathrm{j}2q\omega}\,\mathrm{d}\omega &= \sum_m r[2m]\,\frac{1}{2\pi}\int_{-\pi}^{\pi}\mathrm{e}^{\mathrm{j}2(q-m)\omega}\,\mathrm{d}\omega \\ \delta[q] &= \sum_m r[2m]\delta[q-m] = r[2q]. \end{aligned}$$

We used the orthogonality of the integer complex exponentials, exactly as in Fourier series. This proves both directions:

$$\boxed{\begin{array}{l} \displaystyle\sum_n g_0[n]g_0^*[n-2m] = \delta[m] \quad \text{for every } m \\ \qquad\qquad \Updownarrow \\ |\widehat{g}_0(\omega)|^2 + |\widehat{g}_0(\omega+\pi)|^2 = 2 \quad \text{for every } \omega. \end{array}} \tag{10.10}$$

This is *power complementarity*. It is the squared magnitudes that add to two, not the magnitudes.

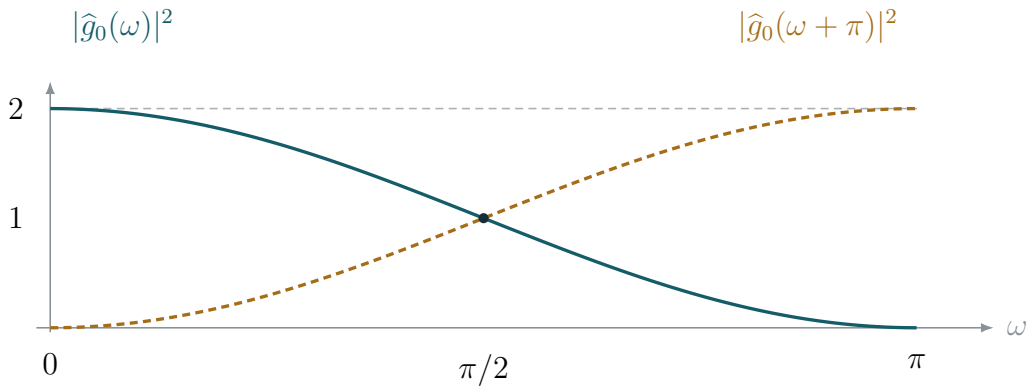


Figure 10.6: For Haar, the responses are $1+\cos\omega$ and $1-\cos\omega$. Their sum is two at every frequency. They both equal one at $\pi/2$.

Orthogonality does not fix the zero-frequency gain. The filter $g_0 = \delta$ has orthonormal even translates and satisfies power complementarity, but $G_0(1) = 1$. The refinement equation supplies an additional constraint.

10.6 Why the low-pass taps sum to $\sqrt{2}$

Orthogonality constrains energy. A low-pass filter intended to refine a scaling function has an additional constraint on its *sum*. We derive that constraint instead of treating the gain $\sqrt{2}$ as an arbitrary convention.

Recall the scaling-function two-scale equation from Lectures 6 and 7:

$$\phi(t) = \sqrt{2} \sum_n g_0[n] \phi(2t - n). \tag{10.11}$$

For the following calculation, assume $\phi \in L^1(\mathbb{R})$ and

$$A := \int_{\mathbb{R}} \phi(t)\, \mathrm{d}t \neq 0.$$

The usual normalization is $A = 1$, but only nonzero A is needed here. Since g_0 is FIR, we may integrate its finite sum term by term:

$$A = \sqrt{2} \sum_n g_0[n] \int_{\mathbb{R}} \phi(2t - n)\, \mathrm{d}t.$$

For each fixed n, set $u = 2t - n$. Then $t = (u + n)/2$ and $\mathrm{d}t = \mathrm{d}u/2$, so

$$\int_{-\infty}^{\infty} \phi(2t - n)\, \mathrm{d}t = \frac{1}{2} \int_{-\infty}^{\infty} \phi(u)\, \mathrm{d}u = \frac{A}{2}.$$

It follows that

$$A = \sqrt{2} \sum_n g_0[n] \frac{A}{2} = \frac{A}{\sqrt{2}} \sum_n g_0[n].$$

Dividing by $A \neq 0$ gives

$$\boxed{G_0(1) = \hat{g}_0(0) = \sum_n g_0[n] = \sqrt{2}.} \tag{10.12}$$

The factor $\sqrt{2}$ in (10.11) normalizes the finer basis functions in energy; the factor $1/2$ comes from their compressed domain.

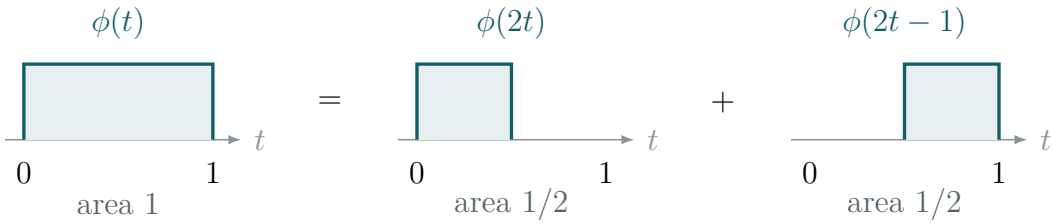


Figure 10.7: Haar makes the integral calculation visible: $\phi(t) = \phi(2t) + \phi(2t-1)$. Here $g_0[0] = g_0[1] = 1/\sqrt{2}$, so the two weights $\sqrt{2} g_0[n]$ equal one.

This is a necessary compatibility condition for the stated refinement model. It does not by itself prove that a given filter generates a scaling function or an MRA.

10.7 The even and odd tap sums are balanced

At zero frequency, (10.10) gives

$$|\hat{g}_0(0)|^2 + |\hat{g}_0(\pi)|^2 = 2.$$

Substitute (10.12): $2 + |\hat{g}_0(\pi)|^2 = 2$. Therefore

$$\hat{g}_0(\pi) = G_0(-1) = 0. \tag{10.13}$$

Here $z = 1$ corresponds to frequency zero, while $z = -1 = \mathrm{e}^{\mathrm{j}\pi}$ corresponds to frequency π.

Separate the even-indexed and odd-indexed taps:

$$E := \sum_m g_0[2m], \qquad O := \sum_m g_0[2m+1].$$

The z-transform splits as

$$G_0(z) = \sum_m g_0[2m] z^{-2m} + \sum_m g_0[2m+1] z^{-(2m+1)}.$$

At $z = 1$, both powers equal one. At $z = -1$, even powers equal one and odd powers equal minus one, also for negative exponents. Hence

$$G_0(1) = \sum_m g_0[2m] + \sum_m g_0[2m+1] = E + O = \sqrt{2},$$
$$G_0(-1) = \sum_m g_0[2m] - \sum_m g_0[2m+1] = E - O = 0.$$

Adding and subtracting these equations gives $2E = \sqrt{2}$ and $2O = \sqrt{2}$:

$$\boxed{\sum_{n \text{ even}} g_0[n] = \sum_{n \text{ odd}} g_0[n] = \frac{1}{\sqrt{2}}.} \tag{10.14}$$

Interpretation using two simple inputs. Let $x_c[n] = 1$ and $x_a[n] = (-1)^n$. These sequences are not in ℓ^2, but FIR convolution is still defined. Direct substitution gives

$$(g_0 * x_c)[n] = \sum_k g_0[k] = \sqrt{2},$$
$$(g_0 * x_a)[n] = \sum_k g_0[k](-1)^{n-k} = (-1)^n \underbrace{\sum_k (-1)^k g_0[k]}_{G_0(-1)=0} = 0.$$

The filter passes a constant with gain $\sqrt{2}$ and removes the alternating pattern. These are endpoint checks, not a complete specification of its passband.

10.8 Conjugate mirror filters: one filter determines the bank

Rather than choose four filters independently, choose one filter with orthonormal even translates. A reversal, conjugation, and alternating signs then supply the second synthesis filter; conjugate reversal supplies the analysis filters.

Theorem 10.3 (Conjugate-mirror construction)**.** Suppose the complex-valued FIR filter g_0 is supported in $\{0, \dots, K\}$, where K is odd, and satisfies

$$\sum_n g_0[n] g_0^*[n - 2m] = \delta[m], \qquad m \in \mathbb{Z}.$$

Define

$$\boxed{g_1[n] = (-1)^n g_0^*[K - n], \qquad h_\ell[n] = g_\ell^*[-n], \quad \ell = 0, 1.} \tag{10.15}$$

Then the resulting two-channel bank is orthogonal and has perfect reconstruction with zero delay.

The filter g_1 is the *conjugate-mirror partner* of g_0. With the power-complementarity condition, this is a pair of *conjugate mirror filters* (CMFs). The rule for forming the partner does not replace the orthogonality condition on the starting filter.

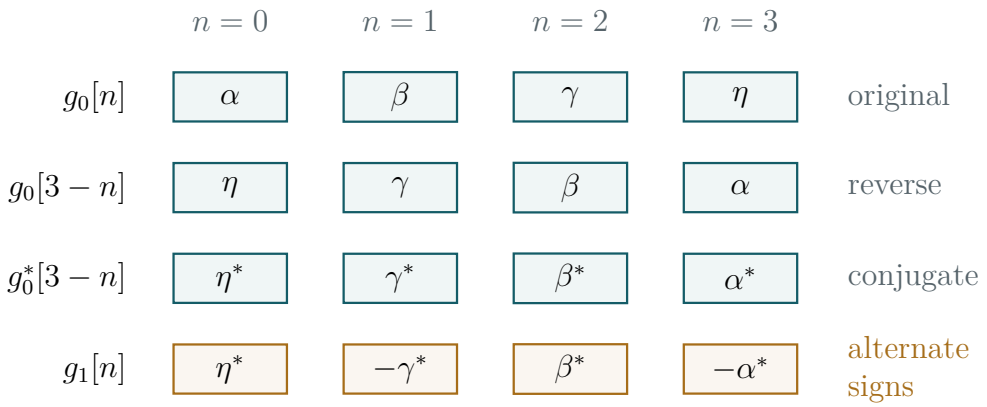


Figure 10.8: The construction for $K = 3$. The symbols illustrate the operations, not a particular choice of filter taps. The starting row must still satisfy double-shift orthogonality.

Both synthesis filters have support contained in $\{0, \dots, K\}$. The analysis filters have support contained in $\{-K, \dots, 0\}$. The zero-delay convention therefore allows noncausal analysis filters; it is not a claim of causal real-time processing without latency.

Proof of the conjugate-mirror construction. We apply theorem 10.2. The g_0 autocorrelation is already assumed. For the other channel, first conjugate the second factor:

$$g_1^*[n-2k] = (-1)^{n-2k} g_0[K-n+2k].$$

Therefore

$$\begin{aligned} r_{11}[k] &= \sum_n (-1)^n g_0^*[K-n]\,(-1)^{n-2k} g_0[K-n+2k] \\ &= \sum_n g_0^*[K-n] g_0[K-n+2k] \\ &= \sum_u g_0[u] g_0^*[u-2k] \qquad (u = K-n+2k) \\ &= r_{00}[k] = \delta[k]. \end{aligned}$$

The signs cancel because $(-1)^n(-1)^{n-2k} = 1$. In particular, the second filter has unit energy.

For the cross-correlation, the conjugation in the inner product cancels the one in the definition of g_1:

$$\begin{aligned} r_{01}[k] &= \sum_n g_0[n] g_1^*[n-2k] \\ &= \sum_n (-1)^n g_0[n] g_0[K+2k-n]. \end{aligned}$$

Neither factor in the last product is conjugated. Set $J = K + 2k$, which is odd, and substitute $u = J - n$:

$$\begin{aligned} r_{01}[k] &= \sum_u (-1)^{J-u} g_0[J-u] g_0[u] \\ &= -\sum_u (-1)^u g_0[u] g_0[J-u] \\ &= -r_{01}[k]. \end{aligned}$$

The second line uses $(-1)^{J-u} = -(-1)^u$. Thus $r_{01}[k] = 0$, and $r_{10}[k] = r_{01}^*[-k] = 0$ as well.

All four double-shift conditions hold. By theorem 10.2, the even translates of the two filters form an orthonormal basis, *including completeness*. Equations (10.4) and (10.5) therefore give exact zero-delay reconstruction and energy preservation. □

The oddness of K is used at a precise point: the indices n and $K + 2k - n$ have opposite parity, so their cross-correlation contributions cancel. For Haar, $K = 1$ and the construction gives $(1, -1)/\sqrt{2}$ from $(1, 1)/\sqrt{2}$.

10.9 Checking reconstruction in the complex z-domain

The basis argument already proves reconstruction for every $x \in \ell^2(\mathbb{Z})$. For the z-domain check, first take a finitely supported input. We verify the perfect-reconstruction identities from Lectures 8 and 9, keeping all conjugations explicit.

Conjugate reversal. Substituting $q = -n$ gives

$$\begin{aligned} H_\ell(z) &= \sum_n g_\ell^*[-n] z^{-n} = \sum_q g_\ell^*[q] z^q \\ &= \left[G_\ell\left(\frac{1}{z^*}\right) \right]^* . \end{aligned}$$

The star outside the brackets conjugates the *entire value*. On the unit circle, $1/z^* = z$, so $H_\ell(\mathrm{e}^{\mathrm{j}\omega}) = G_\ell(\mathrm{e}^{\mathrm{j}\omega})^*$. For complex taps, simply replacing z by z^{-1} would omit the coefficient conjugation.

The conjugate-mirror partner. In the transform of g_1, put $q = K - n$. Since K is odd, $(-1)^{K-q} = -(-1)^q$, and

$$\begin{aligned} G_1(z) &= \sum_n (-1)^n g_0^*[K-n] z^{-n} \\ &= \sum_q (-1)^{K-q} g_0^*[q] z^{-(K-q)} \\ &= -z^{-K} \sum_q (-1)^q g_0^*[q] z^q \\ &= -z^{-K} H_0(-z). \end{aligned}$$

For the second analysis filter, conjugating $g_1[-n]$ gives

$$h_1[n] = g_1^*[-n] = (-1)^n g_0[K+n].$$

Set $q = K + n$ in its transform:

$$\begin{aligned} H_1(z) &= \sum_n (-1)^n g_0[K+n] z^{-n} \\ &= \sum_q (-1)^{q-K} g_0[q] z^{-(q-K)} \\ &= -z^K \sum_q (-1)^q g_0[q] z^{-q} = -z^K G_0(-z). \end{aligned}$$

We have obtained

$$\boxed{G_1(z) = -z^{-K} H_0(-z), \qquad H_1(z) = -z^K G_0(-z).} \tag{10.16}$$

These are finite Laurent-polynomial identities, valid for $z \neq 0$.

Alias cancellation. Replacing z by $-z$ in the second identity of (10.16) gives

$$H_1(-z) = -(-z)^K G_0(z) = z^K G_0(z).$$

Therefore the alias term cancels exactly:

$$\begin{aligned} &G_0(z)H_0(-z) + G_1(z)H_1(-z) \\ &\quad = G_0(z)H_0(-z) - z^{-K}H_0(-z)z^K G_0(z) = 0. \end{aligned}$$

The desired term adds to two. Define $R_0(z) = G_0(z)H_0(z)$. Expanding and grouping by $k = n - m$ gives

$$R_0(z) = \sum_{n,m} g_0[n]g_0^*[m]z^{-(n-m)} = \sum_k r[k]z^{-k}.$$

Thus

$$\begin{aligned} R_0(z) + R_0(-z) &= \sum_k r[k]\big(1 + (-1)^k\big)z^{-k} \\ &= 2\sum_m r[2m]z^{-2m} = 2\sum_m \delta[m]z^{-2m} = 2. \end{aligned}$$

Using (10.16) again,

$$\begin{aligned} G_0(z)H_0(z) + G_1(z)H_1(z) &= R_0(z) + H_0(-z)G_0(-z) \\ &= R_0(z) + R_0(-z) = 2. \end{aligned}$$

In the input–output formula from Lecture 8, these two calculations give

$$Y(z) = \frac{1}{2}(2)X(z) + \frac{1}{2}(0)X(-z) = X(z).$$

The factor two compensates for the factor $1/2$ introduced by the downsampling/upsampling identity.

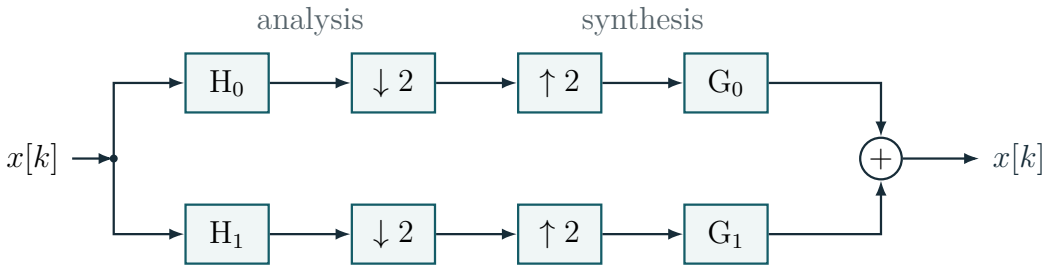


Figure 10.9: The conjugate-mirror bank has zero overall delay in this indexing convention. Both channels are required: their alias terms cancel only when added.

10.10 What does "mirror" mean?

On the unit circle, the first identity in (10.16) reads

$$\widehat{g}_1(\omega) = -\mathrm{e}^{-\mathrm{j}K\omega}\widehat{g}_0(\omega+\pi)^*.$$

Conjugation and the phase factor do not change absolute values. Hence, even for complex-valued filters,

$$\boxed{|\widehat{g}_1(\omega)| = |\widehat{g}_0(\omega+\pi)|.} \tag{10.17}$$

The magnitude response is shifted by half a period. If the low-pass normalization is also imposed, then

$$G_1(1) = -G_0(-1)^* = 0, \qquad G_1(-1) = G_0(1)^* = \sqrt{2}.$$

Thus the partner rejects the constant input and passes the alternating input.

For real taps, a shift becomes a reflection. If g_0 is real,

$$\widehat{g}_0(-\theta) = \sum_n g_0[n]\mathrm{e}^{\mathrm{j}\theta n} = \widehat{g}_0(\theta)^*.$$

Its magnitude is therefore even. Using 2π-periodicity and then this evenness,

$$\begin{aligned}|\widehat{g}_1(\omega)| &= |\widehat{g}_0(\omega+\pi)| = |\widehat{g}_0(\omega-\pi)| \\ &= |\widehat{g}_0(\pi-\omega)|.\end{aligned}$$

On $[0,\pi]$, the transformation $\omega \mapsto \pi-\omega$ reflects a point about $\pi/2$. This is the familiar mirror picture for a real low-pass/high-pass pair.

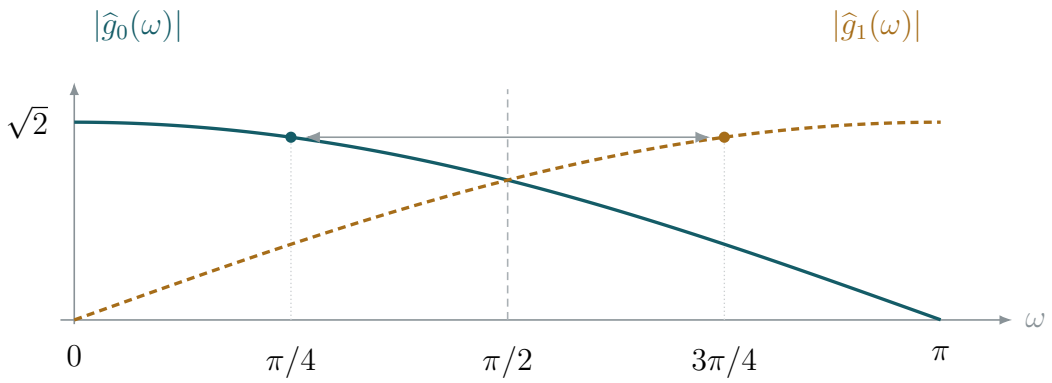


Figure 10.10: Haar magnitudes are $\sqrt{2}\cos(\omega/2)$ and $\sqrt{2}\sin(\omega/2)$ on $[0,\pi]$. Points at equal distances from $\pi/2$ have equal heights on opposite curves.

For complex taps, (10.17) is the general statement; the literal reflection additionally uses real-valued symmetry. The related term *quadrature mirror filter* (QMF) also occurs. Our conjugate-mirror convention is fixed by (10.15).